# Experiment 11

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### Title: Policy Gradient Implementation using REINFORCE Algorithm

### Software Required:

- TensorFlow
- Keras
- Python
- Google Colab

## AIM

To implement the REINFORCE policy gradient algorithm using TensorFlow and Keras for solving a Reinforcement Learning problem.

## THEORY

REINFORCE is a **policy-based Reinforcement Learning algorithm** that directly learns a policy for selecting actions. The policy is represented by a neural network, and its parameters are updated using the rewards obtained from complete episodes. Actions that lead to higher rewards are given greater probability during future decisions.

## PROCEDURE

1. Set up the Python environment in Google Colab.
2. Import TensorFlow, Keras, NumPy and the required libraries.
3. Create a suitable Reinforcement Learning environment.
4. Define the state space and action space.
5. Construct a neural network to represent the policy.
6. Initialize the policy network and its optimizer.
7. Allow the agent to interact with the environment and select actions using the policy.
8. Store the rewards and selected actions for each episode.
9. Calculate the returns and update the policy using the REINFORCE algorithm.

10. Repeat the training process and evaluate the learning performance using the obtained rewards.

# CODE:

```
# EXPERIMENT 11
# Policy Gradient Implementation using REINFORCE Algorithm

!pip -q install gymnasium

import gymnasium as gym
import numpy as np
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers
import matplotlib.pyplot as plt

# ------------------------------------------------
# 1. CREATE ENVIRONMENT
# ------------------------------------------------

env = gym.make("CartPole-v1")

state_size = int(env.observation_space.shape[0])
action_size = int(env.action_space.n)

print("State Size :", state_size)
print("Action Size:", action_size)

# ------------------------------------------------
# 2. CREATE POLICY NETWORK
# ------------------------------------------------

policy = keras.Sequential([
    layers.Input(shape=(state_size,)),
    layers.Dense(32, activation="relu"),
    layers.Dense(32, activation="relu"),
    layers.Dense(action_size, activation="softmax")
])
```

```
optimizer = keras.optimizers.Adam(
    learning_rate=0.001
)

# ------------------------------------------------
# 3. REINFORCE TRAINING
# ------------------------------------------------

episodes = 50
gamma = 0.99

episode_rewards = []

for episode in range(episodes):

    state, info = env.reset()

    states = []
    actions = []
    rewards = []

    done = False
    total_reward = 0

    while not done:

        state_array = np.asarray(
            state,
            dtype=np.float32
        )

        # Get action probabilities
        probabilities = policy(
            state_array.reshape(1, -1),
            training=False
        )[0].numpy()

        # Select action according to policy
        action = np.random.choice(
            action_size,
            p=probabilities
        )
```

```
        # Take action
        next_state, reward, terminated, truncated, info = env.step(action)

        done = terminated or truncated

        # Store experience
        states.append(state_array)
        actions.append(action)
        rewards.append(reward)

        state = next_state
        total_reward += reward

    # ------------------------------------------------
    # 4. CALCULATE DISCOUNTED RETURNS
    # ------------------------------------------------

    returns = []
    discounted_sum = 0

    for reward in reversed(rewards):

        discounted_sum = (
            reward +
            gamma * discounted_sum
        )

        returns.insert(
            0,
            discounted_sum
        )

    returns = np.asarray(
        returns,
        dtype=np.float32
    )

    # Normalize returns
    if len(returns) > 1:
        returns = (
            returns - np.mean(returns)
```

```
    ) / (
        np.std(returns) + 1e-8
    )

states_tensor = tf.convert_to_tensor(
    np.asarray(states),
    dtype=tf.float32
)

actions_tensor = tf.convert_to_tensor(
    actions,
    dtype=tf.int32
)

returns_tensor = tf.convert_to_tensor(
    returns,
    dtype=tf.float32
)

# ------------------------------------------------
# 5. POLICY UPDATE
# ------------------------------------------------

with tf.GradientTape() as tape:

    action_probabilities = policy(
        states_tensor,
        training=True
    )

    selected_probabilities = tf.gather(
        action_probabilities,
        actions_tensor,
        axis=1,
        batch_dims=1
    )

    log_probabilities = tf.math.log(
        selected_probabilities + 1e-8
    )

    loss = -tf.reduce_mean(
        log_probabilities * returns_tensor
```

```
        )

    gradients = tape.gradient(
        loss,
        policy.trainable_variables
    )

    optimizer.apply_gradients(
        zip(
            gradients,
            policy.trainable_variables
        )
    )

    episode_rewards.append(
        total_reward
    )

    print(
        f"Episode {episode + 1:2d}/{episodes} | "
        f"Reward: {int(total_reward):3d}"
    )

# ------------------------------------------------
# 6. CLOSE ENVIRONMENT
# ------------------------------------------------

env.close()

# ------------------------------------------------
# 7. PERFORMANCE GRAPH
# ------------------------------------------------

plt.figure(figsize=(10, 5))

plt.plot(
    episode_rewards,
    marker="o",
    label="REINFORCE Reward"
)

plt.xlabel("Episode")
plt.ylabel("Total Reward")
```

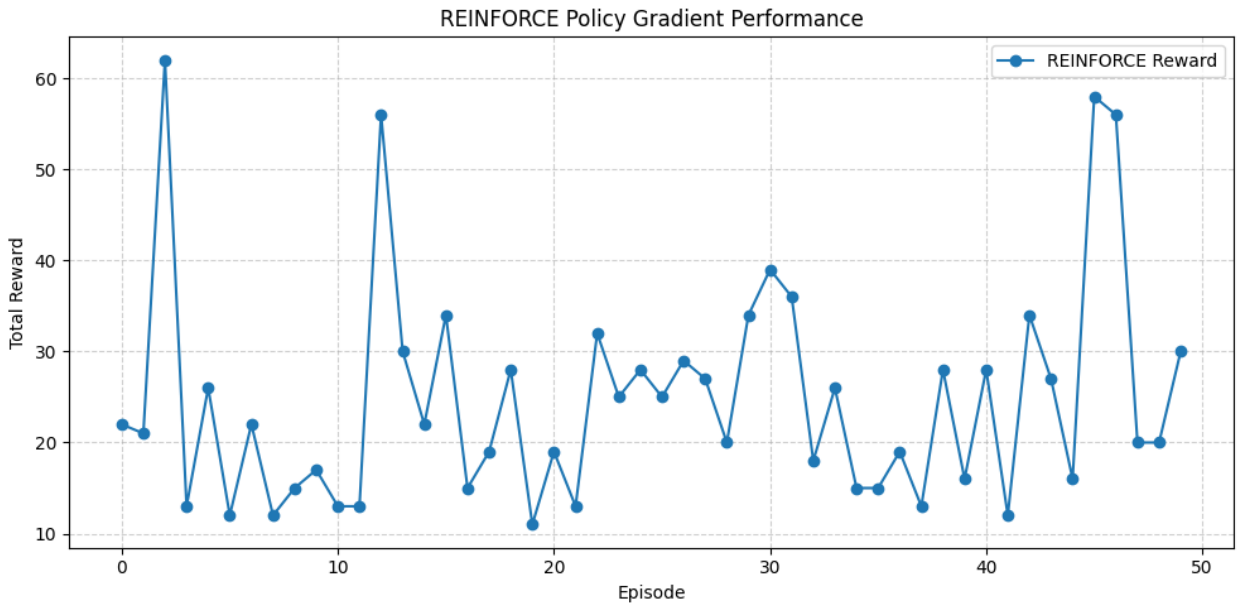
```
plt.title(
    "REINFORCE Policy Gradient Performance"
)

plt.legend()
plt.grid(True, linestyle="--", alpha=0.6)

plt.tight_layout()
plt.show()

# ------------------------------------------------
# 8. RESULT
# ------------------------------------------------

print("\n==================================")
print("REINFORCE TRAINING COMPLETED!")
print("==================================")

print(
    "Best Reward    :",
    max(episode_rewards)
)

print(
    "Average Reward :",
    round(np.mean(episode_rewards), 2)
)
```

## OUTPUT:

```
State Size : 4
Action Size: 2
Episode  1/50 | Reward:  22
Episode  2/50 | Reward:  21
Episode  3/50 | Reward:  62
Episode  4/50 | Reward:  13
Episode  5/50 | Reward:  26
Episode  6/50 | Reward:  12
Episode  7/50 | Reward:  22
```

```
Episode  8/50 | Reward:  12
Episode  9/50 | Reward:  15
Episode 10/50 | Reward:  17
Episode 11/50 | Reward:  13
Episode 12/50 | Reward:  13
Episode 13/50 | Reward:  56
Episode 14/50 | Reward:  30
Episode 15/50 | Reward:  22
Episode 16/50 | Reward:  34
Episode 17/50 | Reward:  15
Episode 18/50 | Reward:  19
Episode 19/50 | Reward:  28
Episode 20/50 | Reward:  11
Episode 21/50 | Reward:  19
Episode 22/50 | Reward:  13
Episode 23/50 | Reward:  32
Episode 24/50 | Reward:  25
Episode 25/50 | Reward:  28
Episode 26/50 | Reward:  25
Episode 27/50 | Reward:  29
Episode 28/50 | Reward:  27
Episode 29/50 | Reward:  20
Episode 30/50 | Reward:  34
Episode 31/50 | Reward:  39
Episode 32/50 | Reward:  36
Episode 33/50 | Reward:  18
Episode 34/50 | Reward:  26
Episode 35/50 | Reward:  15
Episode 36/50 | Reward:  15
Episode 37/50 | Reward:  19
Episode 38/50 | Reward:  13
Episode 39/50 | Reward:  28
Episode 40/50 | Reward:  16
Episode 41/50 | Reward:  28
Episode 42/50 | Reward:  12
Episode 43/50 | Reward:  34
Episode 44/50 | Reward:  27
Episode 45/50 | Reward:  16
Episode 46/50 | Reward:  58
```

```
Episode 47/50 | Reward:  56
Episode 48/50 | Reward:  20
Episode 49/50 | Reward:  20
Episode 50/50 | Reward:  30
```




```
==================================
REINFORCE TRAINING COMPLETED!
==================================
Best Reward    : 62.0
Average Reward : 24.82
```

# RESULT

The **REINFORCE policy gradient algorithm** was successfully implemented, and the agent learned a policy through reward-based updates.